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1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.

Cloud computing is the delivery of computing resources such as servers, storage, databases, networking, and software over the internet. users can access these resources whenever required without owning or maintain the Physical insprastructure.

Four characteristics:

1. On-demand self-service:

Users can create or access compution resources whenever needed without direct assistance from the service Provider.

2. Broad network access:

Cloud services can be accessed through the internet using devices such as laptops, smartphones, and tablets.

3. Resource Pooling:

The provider shares computing resources among multiple users using virtualization and dynamic allocation.

4. Rapid elasticity:

Resources can be increased or decreased quickly according to workload requirements.

2 Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how this change enabled modern cloud platforms.

1. Mainframe computing:

Large centralized computers were shared by many users through terminals.

2. Personal computers:

Computing became available to individual users, but resources were mostly limited to a single machine.

3. client-server computing.

Applications and data were distributed between client computers and centralized servers.

4. Internet evolution:

The growth of the internet allowed users to access applications and resources remotely from anywhere.

5. Virtualization

A single physical server could run multiple virtual machines.

6. Web and internet-based software.

Application began to be delivered through web browsers instead of requiring local installation.

7. Modern cloud platforms:

Cloud provides combined high-speed network, large data centers.

Explain the role of server virtualization in cloud computing. Different between virtualization as an enabling technology and cloud computing as a service model.

Role of server virtualization

Server virtualization divided on physical server into multiple virtual server called virtual maching(VMs). Each VM can run its own operating system and applications.

It helps cloud computing by:

Improving hardware utilization

Reducing infrastructure cost.

Allowing multiple users to share physical resources

Providing isolation between users.

Difference

virtualization	cloud computing
It is a technology	It is a service delivery model
creates virtual resources from physical resources	Provides computing resources to users through a network/internet.
mainly focuses on efficient resource utilization	focuses on providing scalable and on-demand services
Ex: VMware. Creating multiple VMs on one server	Ex: AWS Providing virtual servers through EC2

4. Discuss the role of data centers in delivering cloud services.

A data center is a facility containing servers, storage systems, networking equipment, power systems and cooling infrastructure used to provide computing services.

Data centers play an important role in cloud computing:

1. computing resources: They contain physical servers that provide processing power for cloud applications.

2. Data storage: They store user files, databases, applications

3. Networking: High-speed network equipment connects cloud servers to the internet and users.

4. Availability: Data centers use backup power.

5. security: Physical and digital security mechanisms protect servers and stored data.

6. Scalability: cloud providers can add more servers and storage when increases

5. Differentiate Iaas, Paas, Saas, and Xaas with one suitable real-word example for each model.

Model	Full form	Description	Example
IaaS	Infrastructure as a service	Provides virtual servers, storage and networking	Amazon EC2
PaaS	Platform as a service	Provides a platform and tools for developing and deploying applications.	Google App Engine
SaaS	Software as a service	Provides ready-to-use software through the internet	Google Workspace
XaaS	Anything / Everything as a service	Provides different IT resources or services on a subscription / pay-per-use basis.	Microsoft Azure services,

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